

EDUCATION

Purdue University
Bachelor's Degree in Computer Science

West Lafayette, IN
Expected Graduation : May 2028

EXPERIENCE

Communications & Network Engineer

September 2024 – June 2026

Republic of Korea Navy

ROKS Cheongju, Pyeongtaek, Korea

- Administered network infrastructure of 5 servers and 80 nodes in the ship used by 250+ users across 20 months with 99.97% uptime without vendor support.
- Reduced false-positive IDS alerts by 50% by collaborating with the fleet's cybersecurity team.
- Operated and maintained tactical and satellite communication systems supporting ship-wide data & voice link.
- Built a simple HTML & CSS website in an offline computer to keep track of shipmates' onboard status.

Data Scientist/Engineer

January 2024 – May 2024

ThermoFisher Scientific @ Purdue Data Mine Corporate Partners

West Lafayette Indiana

- Forecasted weekly SKU demand with linear regression over 8 months of shipment history, optimizing bin assignments to minimum volume and cutting required storage footprint by 14%.
- Consolidated 500K+ shipment records from 5 CSV exports into a normalized SQLite schema keyed on package ID, reducing typical query time from 3 minutes to under 5 seconds.
- Raised usable record completeness from 83% to 99% by deduplicating repetitions and correcting false values.

Data Engineer

September 2023 – December 2023

Webee @ Purdue Data Mine Corporate Partners

West Lafayette, IN

- Joined 120K+ sensor readings to 50 stoppage events to construct labeled 10-minute pre-fault windows, enabling supervised fault prediction on previously unlabeled data.
- Built a Flask API and React dashboard visualizing predicted fault probability over live sensor traces.

Teacher & Website Developer

August 2021 – August 2023

Webee @ Purdue Data Mine Corporate Partners

West Lafayette, IN

- Taught coding, engineering, mathematics, and art & technology for 45+ students each summer.
- Proposed and built a website to aid students and faculty, reducing daily inquiries. step-up-two.vercel.app/

Student Cook

September 2023 – May 2024

Purdue Dining & Culinary

West Lafayette, IN

- Prepare and serve food for hundreds of students at Hillenbrand Dining Court in Purdue.

PROJECTS

Personal Server | *TCP/IP, HTTP, Linux, Tailscale*

August 2026 – Present

- Converted a desktop into a personal server to link data across my devices and support my research interests.
- Planning on rewriting server in Rust and add a VMM that boots Linux kernel.
- Upcoming ideas: Balance loads using XDP and add safety with eBPF

Personal Library | *Python, Flask, SQLite, HTML, CSS*

June 2023 – August 2023

- Developed a full-stack web application using with Flask serving a REST API with HTML as the frontend.
- Built a user-created library with author and genre filtering.

TECHNICAL SKILLS

Programming Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS, R

Frameworks: React, Node.js, Flask

Developer Tools: Git, Docker, VS Code, IntelliJ

Libraries: pandas, NumPy, Matplotlib

Systems & Networking Tools : Linux, TCP/IP, SSH, Cisco Routers/Switches

Languages : Korean (First) , English (Professional Working Proficiency), Spanish (High Proficiency)